TESTING THE VIKAS ENGINE

'I can't make the pack smaller, you make the aircraft door big.'

By the time I got back from the ISRO headquarters in a reverse transfer that U R Rao gifted me as soon as he took over as the ISRO chairman in February 1985, work on the integrated test complex at Mahendragiri was in full swing. By the year end we expected to do the first test, if not of a fully-assembled engine or a stage, at least a component of the Vikas engine. But, with the engine incorporated into the PSLV project as part of the second stage, we could not wait for the Mahendragiri facility to get ready to test Vikas.

An option was to test the engine at SEP Vernon. I had discussed this with Dhawan, who was in agreement. U R Rao, the new chairman, was also keen that the engine should be ground-tested at the earliest. Dhawan asked me who other than SEP had the test facility. I did my homework and found DFVLR, a German space agency had the ground test equipment. German scientist Dadieu already had a rapport with India, thanks to his friendship with Sarabhai since the 1960s. We wrote to DFVLR, which said it could do another test, the high altitude test, after we get the engine tested at SEP.

High altitude test is done on the ground with simulated conditions of the upper atmosphere, where the pressure and temperature are different from that of the ground. This was crucial to test the efficacy and the structural integrity of the Vikas engine nozzle, which could behave differently at high altitudes.

Dhawan constituted a committee comprising PSLV project director S Srinivasan, aerodynamics group head T S Prahlad and I to go into the need for a high altitude test. I argued that a high altitude test was avoidable, as it would take more time and money, and it was anyway not necessary because we were using the same nozzle the SEP has been using in Viking. And Viking was already a much tested engine for several Arianne flights.

The others insisted on the test, citing the golden rule in ISRO that no piece of a rocket should be flown without a ground test. Many may still call